Practice Quiz: Perception

Part A: Multiple Choice

1. The goal of state inference in Active Inference is to find $q(s)$ that minimizes: A) Expected Free Energy G B) Variational Free Energy F C) Entropy H D) Surprisal S

2. In `run_state_inference()`, convergence is determined by: A) The VFE reaching zero B) The change in beliefs falling below a threshold C) The posterior matching the prior exactly D) A fixed number of iterations completing

3. If the A-matrix is the identity, observing $o = 2$ in a 3-state system yields posterior: A) $q(s) = [1/3, 1/3, 1/3]$ B) $q(s) = [0, 0, 1]$ C) $q(s) = [0.5, 0.5, 0]$ D) The result depends on the prior

4. `model.log_likelihood(obs)` returns: A) A scalar — the total log-probability of the observation B) A vector — $\ln A[o, :]$ for each state C) A matrix — the full log A-matrix D) The posterior distribution

5. After running `agent.infer_states(obs)`, which of these is updated? A) `agent.q_s` only B) `agent.q_s` and `agent.history["vfe"]` C) `agent.q_s`, `agent.history["vfe"]`, and `agent.history["beliefs"]` D) Only `agent.history`

6. A prediction error of $\varepsilon = [0.15, -0.15]$ after observing $o = 0$ means: A) The agent predicted observation 0 perfectly B) The agent underestimated the probability of observation 0 by 0.15 C) The observation was impossible under the model D) The agent needs to increase γ

7. With a noisy A-matrix ($A = [[0.55, 0.45], [0.45, 0.55]]$) and uniform prior, the posterior after observing $o = 0$ will be: A) Identical to the prior B) Slightly favoring state 0 C) Strongly favoring state 0 D) Deterministic at state 0

Part B: Short Answer

1. Given $A = [[0.8, 0.2], [0.2, 0.8]]$ and prior $q(s) = [0.5, 0.5]$, compute the posterior $q(s \mid o = 0)$ by hand using one iteration of fixed-point inference. Show your work.

2. Explain why `agent.prediction_error(obs).sum()` always equals zero. What mathematical property guarantees this?

3. An agent with a clear A-matrix receives observations $[0, 0, 0, 0, 0]$. After each observation, $q(s_0)$ increases. Explain why there are diminishing returns — why does the 5th observation change beliefs less than the 1st?